



DANILO LEOCATA

Software Engineer · M.Sc. Data Science

CONTACT

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ABOUT ME

Software Engineer with 6+ years of experience designing and scaling robust backend architectures for fintech, gaming, and enterprise clients. Hold an M.Sc. in Data Science (110/110, University of Catania) with a peer-reviewed publication in reinforcement learning (ICAART 2025) and hands-on expertise in PyTorch-based deep learning. Currently driving legacy modernization and microservices migration initiatives in fintech, while actively collaborating on projects that integrate machine learning and AI agents into software systems, continuously expanding expertise in advanced data-driven applications.

EDUCATION

M.Sc. – Data Science

Oct. 2020 – Dec. 2023

🎓 **University of Catania, Italy**

Graduated **110/110**. Specialized in machine learning and data science, complemented by coursework in DevOps methodologies and software engineering. Thesis: *Development of Robotic Navigation Systems in Dynamic Environments based on Deep Reinforcement Learning* — resulted in a peer-reviewed publication at ICAART 2025 (see Publications).

Erasmus+

Sep. 2018 – Jun. 2019

✈ **Technical University of Sofia, Bulgaria**

All exams passed with full marks in English. International exposure across multiple cultures and languages.

B.Sc. – Computer Science

Oct. 2015 – Sep. 2019

🎓 **University of Catania, Italy**

Graduated **100/110**. Thesis: *Electronic Voting Procedures and Safeties*. Core technologies: **C, C++, Java, MySQL, React Native, PHP, JavaScript, HTML, CSS**.

PUBLICATIONS

CrowdSim++: Unifying Crowd Navigation and Obstacle Avoidance

ICAART 2025 — Vol. 2, pp. 533–542 DOI: 10.5220/0013192300003890

M. Rosano, **D. Leocata**, A. Furnari, G.M. Farinella

Proposes a unified RL-based simulator extending CrowdSim to jointly handle crowd navigation and obstacle avoidance in complex environments. Robots equipped with 2D LiDAR and a person sensor learn navigation policies that outperform single-task baselines.

WORK EXPERIENCE

IT Consultant – Full-time

Mar. 2024 – present

 Reply Technology Financial Services, Milano

Fintech Consultant for major banking clients, delivering end-to-end solutions across back-office, and customer-facing platforms from initial requirements and architecture to final production release. Collaborate on the modernization of the bank's core sales platform — which processes **250,000–300,000** daily transactions — helping boost system performance by **50%** during a full-scale migration to AWS. Optimize system reliability through rigorous load testing and monitoring, while managing release pipelines, deployment workflows, and enablement ticketing. By integrating AI agents into daily development workflows, cut delivery time by **30%** and accelerate bug detection by **40%**, strategic time savings reinvested into anticipating future development needs, mastering new frameworks, and pursuing technical certifications.

Java, Spring Boot, Maven, Oracle DB, Kibana, RedHat, JPA, JDBC

Software Engineer – Full-time

Jul. 2022 – Mar. 2024

 Mongibello Tech, Catania

Owned and maintained microservices for game management across national and international gaming platforms operating in 3 markets, processing over 10,000 daily transactions. Delivering end-to-end solutions across multiple concurrent projects, coordinating with cross-country teams to consistently hit tight release deadlines while cutting the defect rate by 50%.

Spring Boot, Gradle, Java, Groovy, MySQL, Bash

FullStack Developer – Part-time

Sep. 2021 – Jul. 2022

 Synapses, Catania

Bug fixing and feature development on the proprietary product; built an internal estimation tool from scratch.

Flutter, Java Spring Boot, Maven, Docker, Kubernetes

FullStack Developer – Full-time

Dec. 2019 – Jun. 2021

 Bizmate, Catania

Developed multiple products from scratch: restaurant reservation web-app, License Manager, and Account Manager. Started with front-end maintenance (**AngularJS, jQuery**), then took full ownership of new builds.

Svelte, NodeJS, Docker, MongoDB, Keycloak, Redis, Bootstrap

FullStack Developer – Training programme

Oct. 2019 – Dec. 2019

 Adecco, Bologna

Adecco partner training programme focused on enterprise Java development.

Java, MySQL, Spring Boot, JPA, JDBC

CERTIFICATIONS

English Proficiency Test – CEFR C2

15 June - 2026
goFLUENT

Assessed listening comprehension, reading comprehension, grammar, and vocabulary. Achieved **C2** level (Jun. 2026).

Assessing Management

24 October - 2024
Corporate Finance Institute

PROJECTS

Selected projects — full list on [GitHub](#).

CrowdSim Real

2023

</> **Python**

Deep RL simulator for LiDAR-based robot navigation in complex, crowded floor plans.

Gamble platform with microservices

2023

</> **Java · Spring Boot**

Microservices-based online football betting platform following distributed systems best practices.

Svelte e-commerce

2022

</> **Svelte · Express**

Full-stack e-commerce demo.

Triplet Trashbin classifier

2022

</> **Python · PyTorch**

Triplet Neural Network for image classification using PyTorch Lightning.

Biobanksprec

2022

</> **Java · Kubernetes · AWS**

Monolith-to-microservices migration with CI/CD (GitHub Actions), Keycloak auth, and AWS deploy.

Football match predictor

2021

</> **Python**

Poisson-based model predicting international football match outcomes.

Trash bin classifier

2021

</> **Python · PyTorch**

CNN classifier for bin fill-level (empty / half / full).

EXTRACURRICULAR

- **English:** Certified B1 Academic level (University of Catania).
- **Italian:** Native proficiency.
- **Spanish:** Conversational proficiency (self-taught, strong listening and speaking skills).

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- **Office Suite:** Advanced proficiency.
 - **Drone Pilot License:** OPEN category A1/A3 (ENAC certified).